



Design and Implementation of Routing Algorithm to Enhance Network Lifetime in WBAN

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Abstract

Today's era is the era of smart and remote applications exploiting advancement in sensors, cloud, Internet of things etc. Major application is in healthcare monitoring and support using wireless body area network (WBAN) in which sensor nodes sense vital physiological parameters and send to server through sink i.e. smart phone nowadays for seamless monitoring. The most significant issue in such applications is energy efficiency which leads to enhanced network life time that ensures uninterrupted seamless services. From source to sink data transmission may occur considering three different scenarios: source to sink single hop direct data transmission irrespective of in-between node distance, source to sink multi hop data transmission in which transmission range of source node is fixed at a threshold to find next forwarder node and transmission range of source node is incremented by affixed value until data gets transmitted to sink. In this work WBAN having different network configurations based on fixed or random positions of nodes have been simulated. Different scenarios with fixed and varying number of nodes are framed and simulated using MATLAB 2020a for performance evaluation of proposed algorithm in terms of energy consumption, network lifetime, path loss etc. due to data transmission from source to sink. Experimental results show that incremental approach is better than direct one in terms of energy consumption, path loss and network lifetime. While selecting transmission range of a source node, it is considered to keep Specific Absorption Rate (SAR) lower to reduce impact on human tissue.

Keywords Healthcare · WBAN · Energy efficiency · Energy consumption · Path loss · Network lifetime · Specific absorption rate (SAR)

1 Introduction

With rapid and successful advancement in sensors, internet of things, smartphone as sensor hub etc., wireless body area network (WBAN) [1–11] is efficiently used in implementing smart and remote healthcare. It helps in seamless health data collection to ensure

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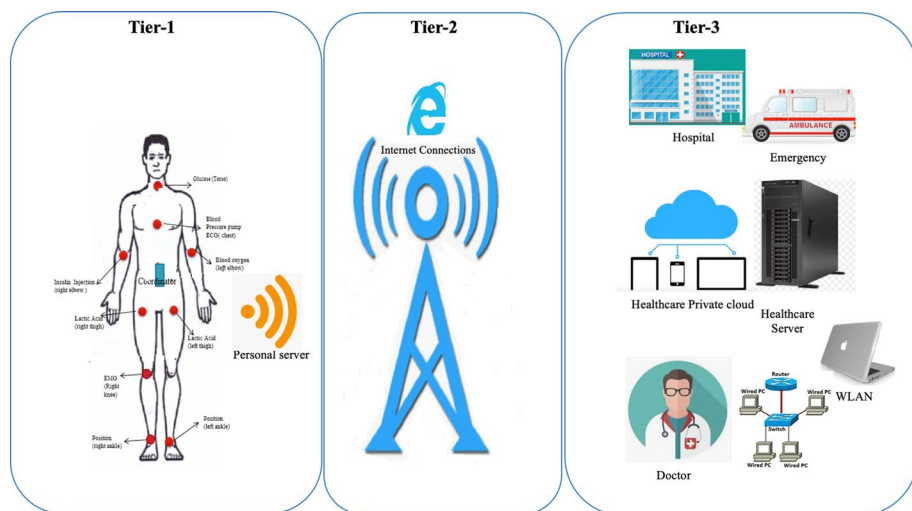


Fig. 1 Basic architecture of WBAN

continuous monitoring and support to elderly people mostly living alone without manual care in current socio-economic scenario. WBAN includes a collection of body sensor nodes that are miniaturized, light weight, ultralow powered devices [5, 6]. It senses vital physiological parameters of human body and transmits to the coordinator or sink and sends the data to the medical server through the internet for diagnosis by the expert to provide necessary advice to the patient in real time. WBAN sensors are equipped with rechargeable energy source and need to be efficiently utilized to enhance network lifetime so that data sensing can go seamlessly. Other important applications in which WBAN is used are sports, military, ambient intelligent, entertainment and machine-to-human or human-to-machine interaction.

Three tier architecture of WBAN based data collection and transmission to the server for processing is shown in Fig. 1. Tier 1: body sensor network formed on human body either by wearable or implantable wireless nodes to coordinator or sink connection e.g. smartphone attached to human body either by Wi-Fi, ZigBee or Bluetooth i.e. any short-range communication, Tier 2: data transmission from coordinator or sink to internet, Tier 3: internet connection to medical server either private or cloud sever based through internet. There are two types of communication or routing in concern: inter and intra WBAN routing.

Intra-WBAN is routing of collected sensor data to the sink or coordinator node, this routing may take place directly from source to sink if within range or through intermediate sensor node or a relay forwarder node for communication at further distance. Node to node or node to sink communication is done by using short range communication technology [6, 8] like IEEE 802.15.1 (Bluetooth), IEEE 802.15.4 (ZigBee) and IEEE 802.15.6 standard which are basically used for implantable sensor node. Mostly Star and Mesh topology have been used in intra WBAN.

WBAN sensor nodes collect vital signals like temperature, blood pressure, EMG and ECG data etc. and send to the coordinator and from coordinator to server using above architecture. As consequence, health hazards like reduced blood flow, enzymatic and tissue damage [7] etc. are observed in many places due to exposure to electromagnetic radiation

of sensor nodes. Specific absorption rate (SAR) [7, 10] is measured in terms of radiation exerted by sensor nodes. However, continuous direct data transmission from the sensor to the coordinator or sink data transmission may increase energy consumption of source node due to long distance transmission and hence network lifetime will be reduced. Multi-hop data transmission through intermediate sensor node would solve this issue. Another factor in WBAN routing is temporal issue which is due to body part movement or whole-body movement that may result into Non-Line of Sight (NLOS) [9, 11] communication among nodes and hence path loss. In particular, energy efficient routing protocol is an important factor for any application developed using WBAN. In Intra body routing, a number of thermal-aware routing protocols have been proposed [2, 9, 12–16]. The works in [2, 16, 17] are for preventing hotspot node in WBAN and in [12] an algorithm is proposed for preventing the hotspot area in a heterogeneous sensor network for improvement of quality of service [18–22]. In this paper, our contribution is to propose and develop a novel routing protocol for WBAN in tier1 combining direct (single hop) and indirect (multi hop) data transmission.

Rest of the paper is organized as follows, Sect. 2 describes a review of the related work with comparison table. Section 3 discusses SAR value analysis related to transmission range of a sensor node and hence energy consumption, Sect. 4 details out methodology with problem identification and proposed work, Sect. 5 illustrates the configuration of the network model followed by simulation setup described in Sects. 6 and 7 show experimental results, finally Sect. 8 concludes the paper.

2 Related Work

In this section, a number of related state-of-the-art works have been described with a comprehensive summary table to have a detailed scenario of current status of this work. Most of the works have designed efficient routing algorithms in resource constrained WBAN and IoT frameworks. Reported works here are compared based on proposed protocol, methodologies applied, performance attributes used for evaluation and simulation or implementation environments.

Huynh et al. [18] proposed a delay-constrained energy efficient cluster based multi-hop routing in WSN protocol. It is a new distributed clustering approach, which determines the best cluster head for each cluster. Here, a new algorithm for energy-cost function and end-to-end delay function for inter cluster routing has been proposed. Author has shown that multi-hop routing algorithm can be used in dissemination of sensor data from cluster heads to a sink with minimum energy cost.

Rashed et al. [19] has proposed an energy-efficient routing protocol in order to enhance the stability period of wireless heterogeneous network. In this proposed scheme, combination of cluster based strategy and chain routing algorithm is used which satisfies both energy and stability of the wireless sensor network in heterogeneous environment.

Ivanov et al. [20] described in his paper Virtual Groups for Patient WBAN Monitoring in Medical Environments for the WBAN. They encountered challenges of the WBAN and monitoring process to suitable quality of data. The authors proposed an architecture which formed virtual group between PDA devices and patients. Author also proposed a new metric, which says to medical expert on quality of received data and is called the Quality of health monitoring.

Monowar et al. [21] proposed a thermal-aware multiconstrained intra-body QoS aware routing protocol for WBAN which is called as TMQoS. Main aim of this protocol is to maintain desired multi-constraints like delay or reliability. Also temperature of the sensors are kept within an acceptable level applying hotspot preventing mechanism to avoid sensor node damage. Proposed TMQoS algorithm selects forwarder nodes for transferring data based on intended QoS demands of desired traffic types.

Ahmad et al. [22] presented a thermal aware routing protocol for application in WBAN. Here fixed wireless sensors node deployment is done according to the nodes energy level. In this protocol for emergency condition data is transferred through single hop, i.e. direct data transfer from source to sink takes place and for normal condition data is transferred through multi-hop i.e. through forwarder node.

Javaid et al. [23] proposed a routing protocol with increased throughput of the WBAN. Author introduced multi-hop mode communication for data transfer from deployment sensor node to the sink through the intermediate nodes. This protocol minimized to the energy consumption, maximized stability period and throughput of the network.

Tanvir et al. [24] has proposed body coordinator node placement strategy, which can increase the network life time. Here authors proposed three different algorithms for placement of body sensor node coordinator. The output result shows that the algorithms along with an appropriate routing protocol is used for energy saving and increased network life time of WBAN up to 47.45% is achieved.

Adhikary et al. [25] has proposed a new routing protocol for WBAN to enhance energy consumption and network life time. In this paper fixed nodes are added to WBAN which act as forwarder node. Fixed node is a proper subset of sensor nodes, they are selected based on cost function metrics. The metrics are transmission power of the nodes, receiver velocity, residual energy and sensor node location with respect to coordinator. This multi-hop routing protocol maximizes network life time and increases packet delivery.

Smail et al. [26] proposed a stable, reliable, energy efficient routing protocol for WBAN. Here author used an objective model to select routing path. The author also shows that state of the art for energy consumption and routing overhead.

Ahmed et al. [27] presented an energy efficient routing protocol based on priority of data. Highest priority is assigned to the emergency data, next is assigned to the on demand data and 3rd priority gets assigned to the periodic data for data transfer from source to sink. Here two non-linear optimization models are introduced for problem formation framework which calculates the best possible performance of WBAN. Proposed models show changes of network life time and throughput in different scenarios.

Ali Khan et al. [28] proposed energy efficient routing protocol for WBAN. Here eight sensors are placed on the human body surface and two sensors are to sense critical data. These two sensors send data directly to the sink. Other six nodes act as forwarder node depending on cost function and minimum distance compared to the entire deployment nodes. This multi-hop routing reduced the energy communication and stable result on network.

Saha et al. [29] has proposed energy efficient routing with and without relay. Nodes in the network are divided into different clusters and relay nodes are placed in cluster head positions selected applying k-means algorithm. Three different scenarios have been described for performance comparison in terms of energy consumptions: Sender node to sink through only relay node, directly from source to sink node and through shortest path only from source to sink node. The scenario of routing through relay node is energy efficient with varying network configurations in terms of number and position of nodes compared to other scenarios.

Ullah et al. [30] presented energy efficient path allocation protocol depending on traffic priority. Authors focus on data transmission delay from source to sink because it may increase packet loss, temperature rise and increase network lifetime. Here designed routing protocol mostly is divided into four folds based on residual energy of the sensor nodes, 1st path allocation is based on priority of type of data like normal, on-demand and emergency, 2nd path allocation strategy is delay aware and energy efficient, where normal data is transmitted through selected optimal and shortest path depending on hotspot generation. 3rd path allocation of data transmission in on-demand algorithm for medical persons like doctors where they can inquire if any emergency or critical condition happened to the patients. 4th path allocation is based on criticality detection algorithm which measured criticalities of vital sign and accordingly allocation of energy efficient and adaptive path is done for priority data thus remove any conflict between them.

Sagar et al. [31] has proposed energy aware health monitoring protocol for Critical Data Routing. Here medical super sensor (MSS) node is a special type of sensor node placed in the human body. This node acts as a controller which managed all the sensor nodes attached to the human body. Sensor node data is compared with pre-defined threshold value and if anything abnormal is found then data transmission process begins. Hence this protocol only cares about the critical data, not considered for normal data, so it is avoiding the redundant picking of normal data. However, this protocol can maximize of energy and network may be alive for longer time.

Zhang et al. [32] proposed energy-aware relay mechanism routing protocol for WBAN. Here authors consider two stages one is initial stage and update stage. In initial phase, coordinator first collects the essential information of each sensor node and in update stage, can invoke enquired method or report method for relay selection in efficient way for routing in the network. Then it can reform the topology according to the network condition. Authors show that proposed protocol improves network lifetime.

Navya and Deepalakshmi [33] proposed an energy efficient routing technique for transmitting critical vital data in wireless body area network under emergency scenarios.

In Table 1 the comparative summary shows that most of the routing algorithms focus on energy efficiency to enhance network lifetime which is very crucial in time critical application like healthcare.

3 Specificabsorption Rate (SAR) Analysis for Energy Consumption

In previous energy efficient routing protocols [2, 4, 5, 8, 12, 13, 18, 19, 23, 25, 29, 34, 35] main components of energy consumption in WBAN are Transceiver unit(Transmit, receive and ideal), sensing unit (sense and actuate) and processing unit (display, aggregate and compute) shown in Fig. 2. In WBAN, there are two components of energy consumption, one is at body network and second one is at system network. In on-body, different heterogeneous sensors that can be wearable or implanted like blood pressure, ECG, EMG etc., are used for collecting vital signals of the patient. So, if SAR value is increased then it may affect the human body blood flow, cause important tissue damage or affect enzymatic reaction. On the system network lifetime and reliability are major concerns. Whereas on communication there is a need to reduce interference, end to end delay and packet delivery ration for increasing throughput and network link reliability, so that it can improve efficient routing algorithm of WBAN.

Table 1 Summary of related works

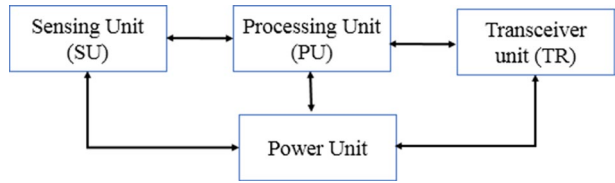
Sl. No	Authors	Year	Proposed work	Methodology	Performance attribute	Simulation/implementation
1	Rashed [19]	2011	Energy efficient routing protocol to enhance network stability	Cluster and chain based approach	Stability period and Network lifetime	MATLAB-7
2	Ivanov [20]	2012	An architecture for group formation and modification is performed with respect to patients' condition and medical officer requirements	A quality of health monitoring metric is used for measuring quality of WBAN data received	Improve Quality of health monitoring	Java based event driven simulator
4	Monowar [21]	2014	Thermal-aware multi-constrained intra-body QoS aware routing	Hotspot preventing mechanism	Energy efficiency	C++
5	Ahmad [22]	2014	Thermal aware energy efficient routing considering both emergency and normal data	Energy efficient routing protocol for increasing network lifetime of WBAN	Network lifetime, packet drops, and throughput	MATLAB
6	Javaid [23]	2015	Routing protocol to enhance the network stability and throughput	Main objective of the routing protocol, sensor data is forwarded to sink in the medium of intermediate nodes using multi hop mode communication	Energy, Network life time, pathloss, throughput	MATLAB
10	Tanvir [24]	2015	Energy efficient routing with effective Body node Coordinator node placement	Main objective of the BNC placement based routing protocol, It is influenced the network life time of the WBAN	Energy, network life- Time	NA
7	Adhikary [25]	2016	Fixed forwarder node is created based on selected cost function metrics	Multi-hop routing	Network lifetime, stability period and throughput	Castalia -3.2
8	Smail [26]	2016	Stable, reliable and energy aware routing	An objective model, select an efficient routing path	Energy consumption and routing overhead	NS-2

Table 1 (continued)

Sl. No	Authors	Year	Proposed work	Methodology	Performance attribute	Simulation/implementation
9	Ahmed [27]	2017	Energy Efficient Routing Algorithm based on data priority	Here three different priorities are set for different types of data transfer from source to sink according to the different conditions and changes of network life time and throughput are optimized	Network throughput, network life time and total energy	MATLAB
11	Ali Khan [28]	2018	Energy efficient multi hop routing protocol for WBAN	Here used multi hop communication to reduced energy and stable network lifetime	Energy, pathloss, network life time	MATLAB
11	Saha [29]	2019	Energy Efficient Routing Protocol using relay or without relay to enhance network lifetime of WBAN	Energy efficient routing protocol	Pathloss, network lifetime	MATLAB
12	Ullah [30]	2019	A traffic priority based four folded delay- aware and energy efficient path allocation routing protocol for WBAN. Different path allocation is done for different scenarios. Energy efficient routing ensures data transfer from source to sink and source to medical persons	Energy efficient routing algorithm for different scenarios like normal data, on-demand data and emergency data	Packet delay, Throughput, Energy consumption,	NS-2

Table 1 (continued)

Sl. No	Authors	Year	Proposed work	Methodology	Performance attribute	Simulation/implementation
13	Kumar Sagar [31]	2020	Energy-Aware critical Data Routing protocol for health monitoring using WBAN	The pre-defined value is compared with new collected sensor value, if there is any change then critical data is sent to the Sink. In this situation only emergency data is sent with priority over redundant normal data	Energy, network lifespan, throughput, efficiency	MATLAB
14	Yu Zhang Bing [32]	2020	The algorithm is applicable in WBAN as it uses an energy aware, adaptive relay mechanism	Network topology gets adjusted adaptively based on energy level of each node so that nodes with lack of energy can be conserved	Residual energy, and lifetime	C++ platform

Fig. 2 Block diagram of a sensor node

According to the previous research, [10, 24, 25, 29, 35–41] researchers say that more energy is consumed in transceiver unit (E_{TR}) compared to the total energy requirement by sensing unit (E_{SU}) and processing unit (E_{PU}). So, $E_{TR} \gg E_{SU} \text{ (AND)} E_{PU}$. Since sensing unit (E_{SU}) and processing unit (E_{PU}) energy is negligible, main focus should be on transceiver unit (E_{TR}) energy consumption. In transceiver unit there are three types of energy: transmission energy (E_{TX}), receiving energy (E_{RX}) and idle listening energy (E_{ID}). Here, E_{TX} consumes more energy rather than E_{RX} and E_{ID} . So, $E_{TX} > E_{RX} \text{ (OR)} E_{ID}$ and energy consumption of receiver is approximately equivalent to the idle energy consumption, Thus $E_{RX} \approx E_{ID}$. Therefore depending upon the transceiver energy (E_{TR}), all the previous routing algorithm has described their energy model. In WBAN sensors are either implanted or wearable on the human body. Thus, the body absorbs Electromagnetic Radio Frequency (EMRF) waves radiated from sensor nodes when they are active during data transmission. As the transmission range increases thus the transceiver energy increases causing higher SAR value. Higher SAR value can damage human body tissue. On the other hand, low residual energy affects the attenuation and interference, hence resulting in low link reliability. In WBAN sensor nodes directly send data to the coordinator, which can increase SAR value. Therefore any imprecise approach can be cause health hazards. So there is a need to quantify SAR value of the node for its various activities in the network. Energy consumed per unit mass per unit time is called as Specific Absorption Rate (SAR). If M is the mass and t is the time then SAR is as given in Eq. (1)

$$SAR \approx \frac{E_{consum}}{M \times t} \quad (1)$$

Equation (1) can be re-written as:

$$SAR = \frac{1}{M} \times \frac{dE_{consum}}{dt} \quad (2)$$

where $\frac{dE_{consum}}{dt}$ is energy consumption rate. According to energy consumption components of a sensor node described earlier, we can re-write SAR as follows:

$$SAR = \frac{1}{M} \times \left(\frac{dE_{TR}}{dt} + \frac{dE_{SU}}{dt} + \frac{dE_{PU}}{dt} \right) \quad (3)$$

Since, $E_{TR} \gg E_{SU} \text{ (AND)} E_{PU}$

$$SAR = \frac{1}{M} \times \left(\frac{dE_{TR}}{dt} \right) \quad (4)$$

$$SAR \cong \frac{1}{M} \times \left(\frac{dE_{TX}}{dt} + \frac{dE_{RX}}{dt} \right) \quad (5)$$

Hence $E_{RX} \approx E_{ID}$

Table 2 Sample of specific absorption rate (SAR) [38]

Whole-body	Partial-body	Hands, wrists, feet, ankles
0.08	1.6	4.0

Table 3 Relationship between input power and SAR [40]

Input power (W)	SAR (W/Kg)
0.11	1
0.54	5
1.1	10

However, SAR of WBAN sensor node can be described in above Eq. (5), energy consumption due to sensing, processing and forwarding towards the destination. In case of WBAN, receiving energy (E_{RX}) of a sensor node is minimum, so it can be negligible. So Eq. (5) can be re-written as Eq. (6):

$$SAR = \frac{1}{M} \times \left(\frac{dE_{TX}}{dt} \right) \quad (6)$$

Here $\frac{1}{M}$ is a constant and denoted as ∂ , Eq. (6) is rewritten as given below:

$$SAR = \partial \times \left(\frac{dE_{TX}}{dt} \right) \quad (7)$$

Then, if E_p units of energy is consumed to transmit a single bit of data, then transmission of k bit data would cause $SAR \cong \partial \times k \times \left(\frac{dE_p}{dt} \right)$ or

$$SAR = \partial \times k \times \text{transmission power} \quad (8)$$

Portable device like mobile, camera, sensor etc. with multiple radius of radio range, each radius can run simultaneously and expose the user to some level of electromagnetic radiation. In case of body human tissue, researchers give some SAR value [38] as shown in Table 2.

To increase the coverage area of portable devices, increasing power may cause potential health risks through exposure to radio frequency (RF) fields. Table 3 shows how SAR increases with increase in input power of portable devices:

However, if we increase the input power then energy consumption of the sensor node increases. Hence, if the body temperature rises by 1–2 °C then a large number of physiological effects have been characterized in human tissue. With temperature changes in the various organs or tissues of the body [10, 41], it becomes difficult to measure vital signals correctly using sensor nodes of WBAN.

4 Methodology

4.1 Problem Identification

In previous research [29] we have seen that, using relay node placement in the body network, eminently increases the longevity of the network life time and throughput of the

body area network, to understand its impact [29] we have used the same system model to evaluate the energy efficient routing using relay node and not using relay node in medical body sensor network. While designing routing in WBAN for healthcare, energy efficiency to increase network lifetime and SAR to be considered while transmitting signal from source to sink as SAR is related to transmission power which has to be increased with higher transmission distance. In this paper different routing scenarios are designed with varied transmission distance of sensor nodes to analyze energy efficiency and network lifetime while keeping SAR into consideration. Since sensors are implanted or wearable, increased SAR value may damage human body tissue.

4.2 Proposed Algorithm

A number of heterogeneous physiological sensor nodes are placed at fixed position in reference to that in [29, 35, 42] as shown in Fig. 3a. Sensor nodes are randomly placed on human body as shown in Fig. 4a.

Two different routing algorithms are proposed here based on different routing strategy. Algorithm 1 is based on fixed distance threshold and Algorithm 2 is based on incremental distance threshold-based routing.

4.2.1 Algorithm Description

In Algorithm 1 nodes are placed in fixed positions and distance from nodes to coordinator node is calculated. Fixed distance threshold is set as 0.5 cm and the sender node finds for next forwarder node within coverage area within 0.5 cm radius. If multiple

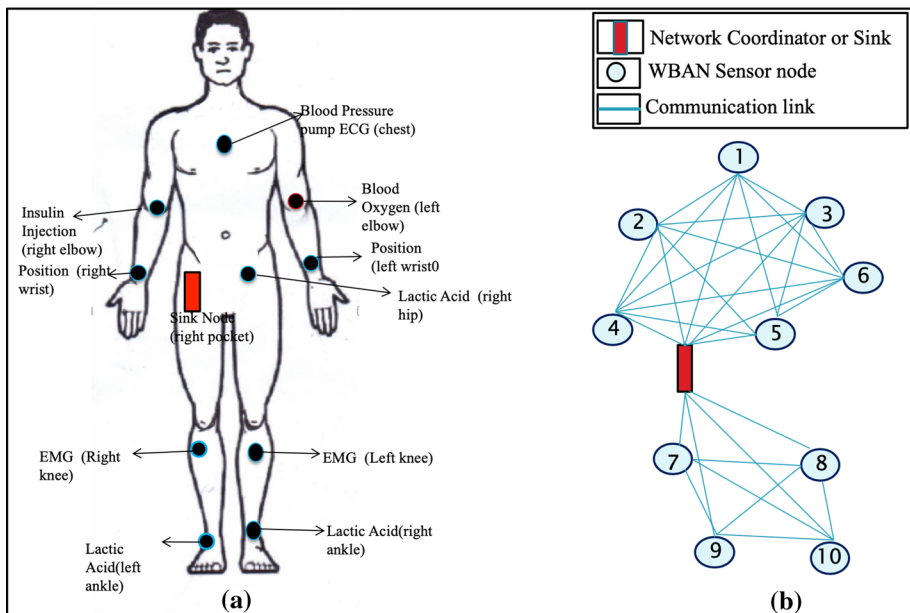


Fig. 3 a Fixed placed heterogeneous node positions on human body, b example network connection

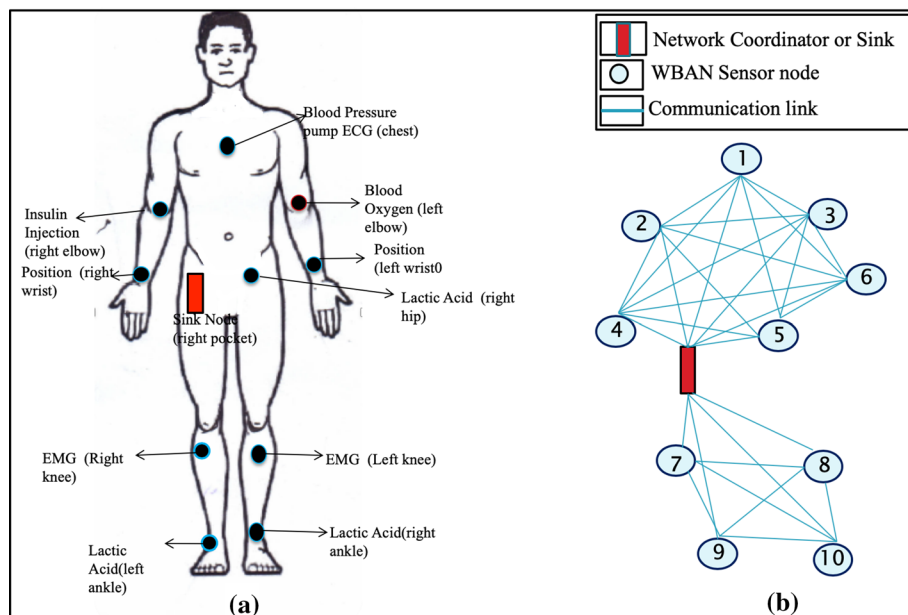


Fig. 4 **a** Randomly placed heterogeneous node positions on human body, **b** example of network connection

nodes are found then the node having minimum distance from the coordinator node is selected. The node with minimum distance and not recently acted as forwarder node is selected in case of multiple such nodes. This process continues until the data reaches to the coordinator. Thus the data forwarding path from sender node to coordinator node gets established and a HELLO message is sent as an acknowledgement.

In Algorithm 2, the distance threshold is reduced than that in Algorithm 1 for performance evaluation purpose. In algorithm 2, the initial distance threshold (Th) is set as 0.1 cm. The next forwarder node in the coverage area with 0.1 cm radius is found and the same process iterates until the coordinator node is reached. If more than one node in the coverage area contests as a probable forwarder node, then similar to algorithm 1, minimum distance from coordinator node to the contestant nodes is held to be the deciding parameter for selection of the forwarder node. A routing path to transfer this data is established and a HELLO message is sent as an acknowledgement. But if within 0.1 cm radius coverage area there is an absence of a forwarder node, then initial distance threshold Th is increased incrementally by 0.1 cm ($Th + 0.1$). In the newly selected coverage area, the next forwarder node is identified, as previously discussed and yet again if any node is not found, the Th value is increased again by 0.1 cm. This procedure continues until the coordinator node in the coverage area is finally reached.

In the first algorithm and other previous proposed algorithms, we see that number of hop count is less than algorithm 2 and this makes the protocol more efficient with respect to energy, path loss, network lifetime. This network characteristic makes significant effect of the Specific Absorption rate (SAR) in the Sect. 7 discuss about it.

Algorithm 1: Fixed Threshold based routing protocol

```

1. Initialization :  $N_i = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ 
2. Initialization Sink Position ( $N_{x_1} N_{y_1}$ ).
3. Place different body sensor node in the different position;
4. Sink node is placed/ selected within a WBAN and place it position at ( $N_{x_1} N_{y_1}$ ).
5. After that, assign initial energy  $E_0$  to every nodes ( $N_i$ );
6. Calculate distance ( $d$ ) every nodes ( $X_i, Y_i$ ) from Sink node ( $N_{x_1} N_{y_1}$ ).
for i = 1 to N
    
$$d = \sqrt{(N_{x_1} - x_1)^2 + (N_{y_1} - y_1)^2} \dots \sqrt{(N_{x_1} - x_N)^2 + (N_{y_1} - y_N)^2}$$

    Here, ( $N_{x_1} N_{y_1}$ ) = Sink node position and ( $x_i, y_i$ ) = every sensor node position.
    end for
7. Node  $N_i$  want to send data Sink( $S_{x_1}, S_{y_1}$ )
8. Set  $Th = 0.5$  around  $N_i$  // coverage area is with 0.5 cm radius around  $N_i$ 
9. if ( $N_i \leq 0.5$  and find Sink node in  $Th$  range )
    Forward data directly to the Sink
10. else
11. for i =  $N_i$  to Sink Node
12. for i = 1 to M
    if ( $Node(i) \leq 0.5$ )
        
$$R(i) = \sqrt{(M_{x_1} - x_1)^2 + (M_{y_1} - y_1)^2} \dots \sqrt{(M_{x_1} - x_M)^2 + (M_{y_1} - y_M)^2}$$

        Here, ( $M_{x_1}, M_{y_1}$ ) = within range sensor node position and ( $x_i, y_i$ ) = every sensor node position.
    13. end if
    14. end for
15. sort R in descending order
16. for j = 1 to N
17. if ( $R[j] \neq$  Sink nodes address)
18. if ( $R[j] \leq Th$  and nearest to Sink and not visited )
19.  $N_j$  Set as a forwarder
20. end if
21. else
22.  $N_j$  Set as a forwarder
23. end if
24. end for
25. end for
26. Send HELLO message and Set routing path
27. else
28. Network in ideal state or dead
29. end if
30. end

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Algorithm 2: Incremental threshold based routing protocol

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1. Initialization :  $N_i = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ 
2. Initialization of Sink Position ( $N_{x_1}, N_{y_1}$ ).
3. Place different body sensor node in the different positions;
4. Sink node is placed/ selected within a WBAN and place it position at ( $N_{x_1}, N_{y_1}$ ).
5. After that, assign initial energy  $E_0$  to every node ( $N_i$ );
6. Calculate distance( $d$ ) of every node ( $x_i, y_i$ ) from Sink node ( $N_{x_1}, N_{y_1}$ ).

    for  $i = 1$  to  $N$ 
         $d = \sqrt{(N_{x_1} - x_1)^2 + (N_{y_1} - y_1)^2} \dots \sqrt{(N_{x_1} - x_N)^2 + (N_{y_1} - y_N)^2}$ 
    Here, ( $M_{x_1}, M_{y_1}$ ) = Sink node position and ( $x_i, y_i$ ) = every sensor node position.
end for

7. Node  $N_i$  wants to send data Sink( $S_{x_1}, S_{y_1}$ )
8.  $Th = 0.1$  around  $N_i$  // coverage area is with 0.1 cm radius around  $N_i$ 
9. if ( $N_i \leq Th$  and find Sink node in threshold value )
    Forward data directly to the Sink
10. else
11. for  $N_i$  to Sink Node
12. if ( $N_i \leq Th$  )
13. for  $i = 1$  to  $M$ 
    if ( $N_i(i) \leq Th$  )
         $R(i) = \sqrt{(M_{x_1} - x_1)^2 + (M_{y_1} - y_1)^2} \dots \sqrt{(M_{x_1} - x_M)^2 + (M_{y_1} - y_M)^2}$ 
        Here, ( $N_{x_1}, N_{y_1}$ ) = Within range sensor node position and ( $x_i, y_i$ ) = every sensor node position.
14. end for
15. sort  $R$  in descending order
16. for  $j = 1$  to  $N$ 
17. if ( $R[j] \neq \text{Sink nodes address}$ )
18. if ( $R[j] \leq Th$  and nearest to Sink and not visited )
19.  $N_j$  is being set as a forwarder node
20. end if
21. else
22.  $N_j$  Set as a forwarder node
23. end if; end for; end if
24.  $Th = Th + 0.1$ 
25. end for
26. Send HELLO message and Set routing path
27. end

```

5 Network Model

WBAN is an advanced network which provides seamless routing to ensure energy efficiency and enhanced network lifetime. There is a fixed position for every sensor node in WBAN with cognitive radio interference and it operates with limited energy sources. In Figs. 3a, 4a different heterogeneous sensor nodes are shown. We considered the following assumptions:

- I. All sensor nodes have fixed positions with similar significances and capabilities but used in different purposes.
- II. All sensor nodes have their own energy sources. Energy is consumed according to the distance traversed by the data sent from the sensor nodes.
- III. Radio signals or Electromagnetic radio frequency (EMRF) signals have identical energy attenuation in all directions.
- IV. To keep SAR low, either in single hop or multi hop transmission range of a sensor node is fixed at a threshold, if within range no node is there then range is increased incrementally by equal value unless next forwarder node is found.

- V. All heterogeneous nodes are capable of sensing vital signals from the human body and forwarding it to the coordinator.

5.1 Energy Model

Shown in Fig. 5 [29, 35] a simplified model for radio hardware energy dissipation. Transmitter energy dissipates to run the radio electronics and power amplifier. The receiver energy dissipates to run the radio electronics. In the system receiver received k -bit data, then energy spent for the radio system as Eq. (9).

$$E_{RX}(k) = E_{elec} \times k \quad (9)$$

Here E_{elec} is the energy consumption factor, it is dependent on factors such as digital coding, modulation, filtering and spreading of the signal.

The radio hardware system transmitting k —bit data over a distance, then the energy consumption of the radio system is given as Eq. (10).

$$E_{TX}(k, d) = E_{Tx-elec}(k) + E_{TX-amp}(k, d) \quad (10)$$

$$E_{TX}(k, d) = \begin{cases} k \times E_{elec} + k \times \epsilon_{fs} \times d^2, & \text{if } d < d_0 \\ k \times E_{elec} + k \times \epsilon_{mp} \times d^4, & \text{if } d \geq d_0 \end{cases} \quad (11)$$

Here, E_{elec} is the electronics energy consumption factor, amplifiers ϵ_{fs} and ϵ_{mp} are required to maintain for acceptable SNR and the threshold distance between transmitter and receiver denoted by d_0 , where $d_0 = \sqrt{\frac{\epsilon_{fs}}{\epsilon_{mp}}}$.

For the experimental purpose, described here, power loss for the single hop communication is d^2 and multi-hop is d^4 channel models in the Eq. 11, which depends on routing path between transmitter and receiver. In the experiment, different parameters used are shown in Table 4 [15, 33, 42, 43].

5.2 Network Configurations

For our experimental purpose we made two network configuration scenarios—1) NC1: Network configurations having fixed node positions on human body [29, 35]. Different nodes are placed at different body parts such as left and right ankle, left and right wrist, left and right elbow, left and right knee, at fixed node positions in reference to Fig. 3a, hence number of nodes is fixed and 10. Other one is NC2: Network Configuration 2 having

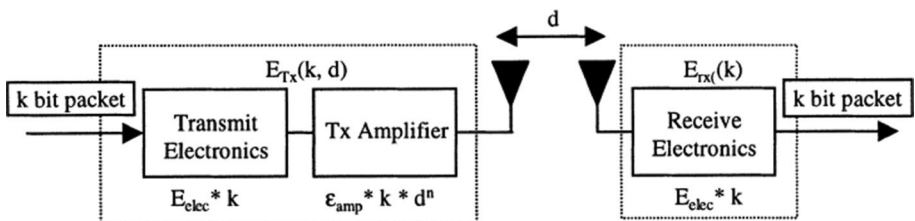


Fig. 5 Radio energy dissipation model

Table 4 Parameters and values

Parameters	Value nRF2401	Value CC2400
Amplifier energy consumption in transmitter sensor (E_{amp})	1.97×10^{-9} J/bit	271 J/bit
Initial energy of the sensor (E_{int})	5 J	5 J
Packet size (k)	2400 bit	2400 bit
Data aggregation	5nJ	5nJ
Transmission energy (E_{TXelec})	16.7 nJ/bit	96.9 nJ/bit
Receiving energy (E_{RXelec})	36.1 nJ/bit	172.8 nJ/bit

random node positioning on human body. Here sink or coordinator is placed in right pocket as shown in Figs. 3a, 4a respectively. In this Network Configuration number of nodes are varied in three different scenarios as 10, 15, 20 and are randomly placed in each scenario. Sensor nodes send data directly to the sink or coordinator if sensing node is in proximity otherwise sensing nodes send data to the forwarder node and forwarder node in turn sends data to the coordinator or sink in multi-hop. To see performance of our proposed protocol in terms of network lifetime, number nodes have been varied and placed randomly in NC2.

6 Simulation Setup

Simulation has been performed to evaluate the performance of proposed algorithm against predefined algorithms like source to sink direct data transfer and data transfer through shortest path algorithm. MATLAB 2020a tool was used for simulation and performance evaluation of different metric of the network.

In Network Configuration 1 (NC1) 10 physiological sensors are placed in according to reference Fig. 3a [29, 35, 42] sensor nodes and coordinator node are placed as per x-axis and y-axis coordinates as given in Table 5.

For simulation in Network Configuration 1, 10 different nodes are placed in the different positions of the body surface and coordinator is placed on right hip of the body, and considered X-axis coordinate as 30 and Y-axis coordinate as -5 . In Simulation environment, setup area of $100 \text{ cm} \times 100 \text{ cm}$ is considered [29] for node placement in MATLAB as shown in Fig. 6.

Network Configuration 2-Scenario 1: (NC2-SC1)—physiological sensors are placed in random way on the human body surface shown in Fig. 4a, sensor nodes and coordinator node are placed as per x-axis and y-axis coordinates as given in Table 6.

In simulation Network Configuration 2, 10 different nodes are randomly placed in the different positions of the human body surface and coordinator is placed in right hip of the body and considered X-axis as 30 and Y-axis as -5 respectively in simulation environment considering setup area as $100 \text{ cm} \times 100 \text{ cm}$ [29]. The placed nodes in MATLAB area are shown in Fig. 7.

Network Configuration 2-Scenario 2: (NC2-SC2)—physiological sensor nodes are increased to 15 and are placed in random way on the human body surface, sensor nodes and coordinator node are placed as per x-axis and y-axis coordinates as given in Table 7.

In Network Configuration 2-Scenario 1: (NC2-SC1)—15 different nodes are randomly placed in the different positions of the human body and the coordinator is placed

Table 5 Sensor node positions (XY-coordinates)

Sensor	Coordinate-X	Coordinate-Y
Node 1	20	35
Node 2	60	35
Node3	20	10
Node 4	60	10
Node 5	45	45
Node 6	50	- 10
Node 7	35	- 30
Node 8	45	- 30
Node 9	35	- 60
Node 10	45	- 60
Sink or Coordinator	30	- 5

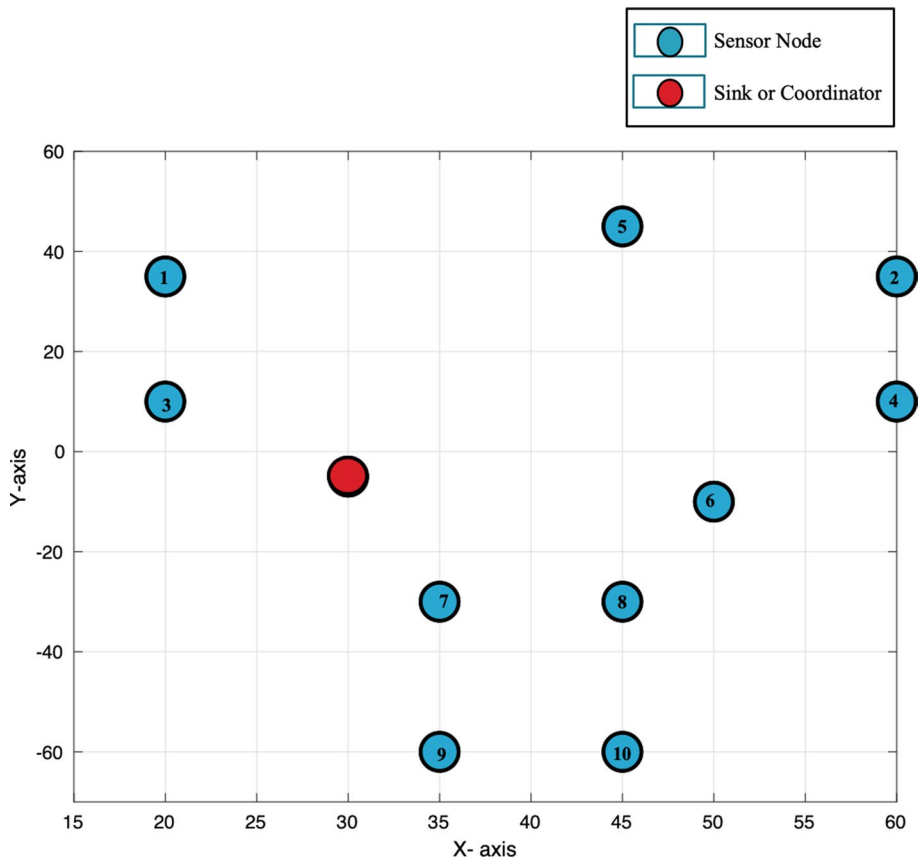
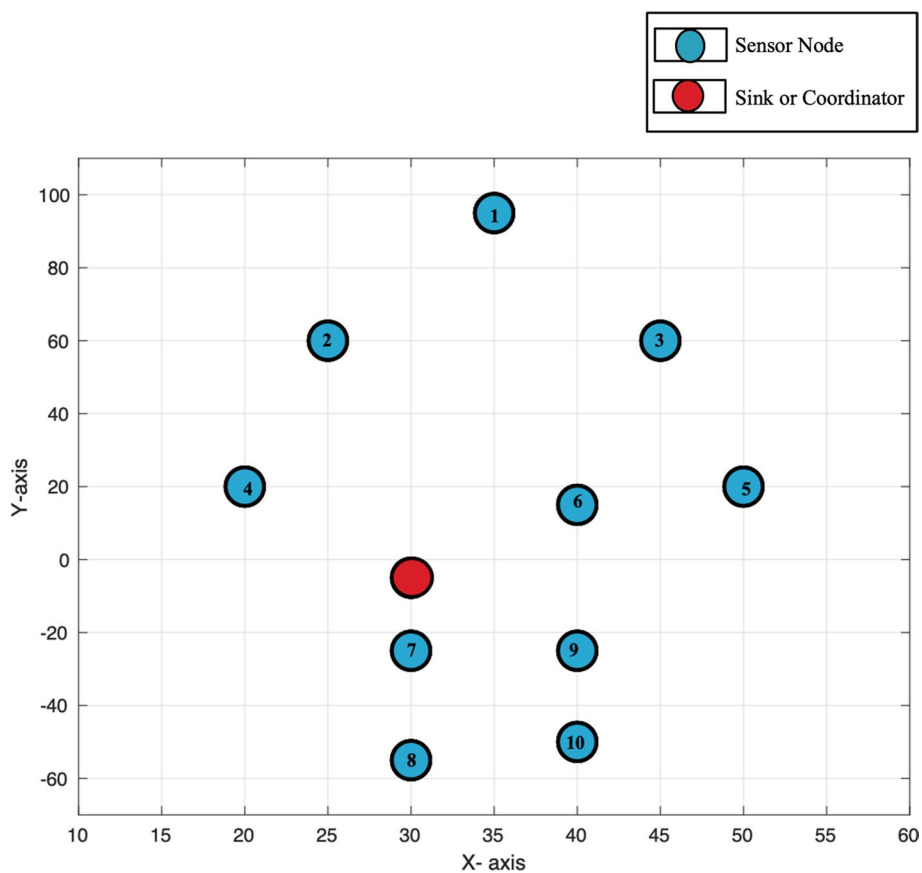
**Fig. 6** Node deployment in MATLAB area

Table 6 Random placed 10
Sensor nodes (XY-coordinates)

Sensor	Coordinate-X	Coordinate-Y
Node 1	35	95
Node 2	25	60
Node3	45	60
Node 4	20	20
Node 5	50	20
Node 6	40	15
Node 7	30	− 25
Node 8	30	− 55
Node 9	40	− 25
Node 10	40	− 50
Sink or coordinator	30	− 5

**Fig. 7** Node deployment in MATLAB area (10 nodes)

on right hip of the body, considered X-axis as 30 and Y-axis as -5 respectively in simulation environment considering setup area as $100\text{ cm} \times 100\text{ cm}$ [29]. The placed nodes in MATLAB area as mentioned above are shown in Fig. 8.

In Network Configuration 2-Scenario 3: (NC2-SC3)—number of physiological sensors is increased to 20 and are placed in random way on the human body surface, sensor nodes and coordinator node are placed as per x-axis and y-axis coordinates as given in Table 8.

In Network Configuration 2-Scenario 3: (NC2-SC3)—20 different nodes are randomly placed in the different positions of the human body surface and coordinator is placed on right hip of the body, considered as X-axis=30 and Y-axis= -5 respectively, shown in red in Fig. 9. Simulation environment is considering setup area as $100\text{ cm} \times 100\text{ cm}$ [29]. The placed nodes in MATLAB area are shown in Fig. 8. Reference distance or threshold distance d_0 is considered as 0.1 cm [2, 29, 35, 39] for all Network Configurations.

In a medical e-healthcare system, different body sensors are placed to acquire patient health vitals. A relative study of nRF 2401 [43] and chipcon CC 2420 [43] transceiver is done to find nRF 2401 better compared to chipcon CC 2420 transceiver in terms of energy consumption and packet delivery ratio.

7 Experimental Results

Simulation using MATLAB2020a is performed to evaluate proposed fixed and incremental threshold based algorithms against routing algorithm in which direct data transfer takes place from source to coordinator. Body sensor nodes and coordinator are placed in different X and Y coordinates of X-axis and Y-axis in different Network Configurations as shown in previous section.

Table 7 Randomly placed 15 Sensor nodes (XY-coordinates)

Sensor	Coordinate-X	Coordinate-Y
Node 1	35	95
Node 2	25	60
Node3	45	60
Node 4	20	20
Node 5	50	20
Node 6	40	15
Node 7	32	70
Node 8	37	70
Node 9	35	30
Node 10	30	40
Node 11	40	45
Node 12	30	-25
Node 13	30	-55
Node 14	40	-25
Node 15	40	-50
Sink or coordinator	30	-5

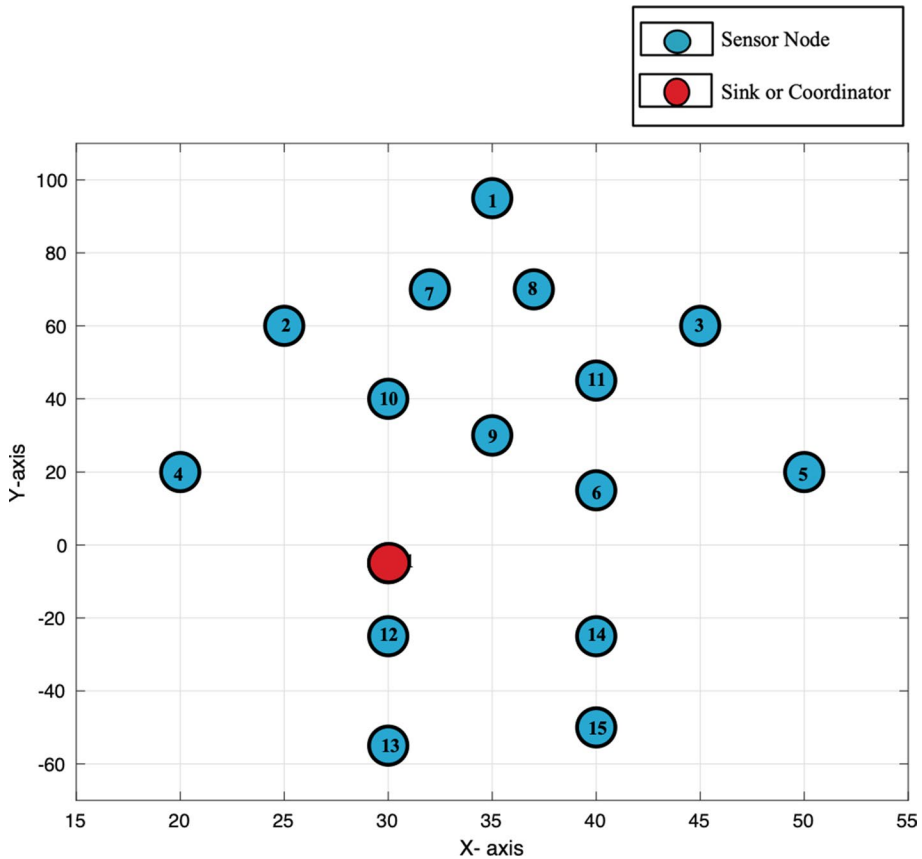


Fig. 8 Node deployment in MATLAB area (15 nodes)

7.1 Energy Performance Metric

Figures 10, 11, 12 and 13 respectively show the comparison among the energy consumptions in different node placement network configurations i.e. NC1 with fixed nodes and NC2 with varying nodes respectively.

In all network configurations, sink node position at ($X=30, -5$) is fixed. Proposed new fixed threshold based model and incremental threshold based model play important roles in minimizing energy consumption among different scenarios. In the earlier proposed model source node directly sends data to sink node in single hop as shown in red and black in Fig. 10. Using shortest path through intermediate relay node, either in single hop or multihop as shown in Fig. 10. Source to sink direct and shortest path are represented in red and black respectively. Blue and green show fixed threshold and incremental threshold based model in Figs. 10, 11, 12, 13. In the new algorithm, the radius of coverage area of a source is fixed at a threshold of 0.5 cm (because corresponding SAR value of this transmission range is not harmful for human tissue) represented in blue and second incremental threshold model is represented in green.

Table 8 Randomly placed 20 sensor nodes (XY-coordinates)

Sensor	Coordinate-X	Coordinate-Y
Node 1	35	95
Node 2	25	60
Node3	45	60
Node 4	20	20
Node 5	25	20
Node 6	50	20
Node 7	45	20
Node 8	40	15
Node 9	30	20
Node 10	32	70
Node 11	37	70
Node 12	35	30
Node 13	30	40
Node 14	35	50
Node 15	40	45
Node 16	30	– 25
Node 17	30	– 55
Node 18	40	– 25
Node 19	40	– 50
Node 20	35	– 5
Sink or coordinator	30	– 5

Here two scenarios are framed: fixed threshold based and incremental threshold based scenarios. Both are compared directly with source to sink direct protocol and shortest path protocol. In fixed threshold based algorithm, source node finds next forwarder node within coverage area with radius equal to fixed threshold value, here fixed at 0.5 cm. In this scenario, the node which is farthest in the range having radius of 0.5 cm around the sender node and nearest to the sink acts as forwarder node. This node transmits its data to the sink, thus data transmission takes place in multi-hop.

In Fig. 10 it is shown that node 1 to node 5 are at far distances from sink. In that situation data is sent in multi hop, but node 6 to node 10 are nearer to sink, so data is sent in single hop. Hence, in Shortest path algorithm some sender node sends data to the sink using single hop or multi hop, for example node 1 sends data to the sink using single hop and node 2 sends data to the sink using multi hop. Here node 2 takes more distance in terms of hop-count compared to the others protocols while energy consumption for node 2 is minimum compared to the other three models.

In incremental threshold model, source node first sets distance threshold value 0.1 cm and searches for the next forwarder node within the coverage area having radius 0.1 cm. If any node is within the coverage area having radius 0.1 cm, then this node is selected as the forwarder node and the data is forwarded to this node. If no suitable node is found within the range then the threshold is increased in steps of 0.1 cm until next forwarder node is found. In some cases more than one node can be found within the range but the node which is farthest from the source node and nearest to the sink is selected as the next forwarder node. Figure 10 shows that node 9 and 10 are nearest to the sink and directly

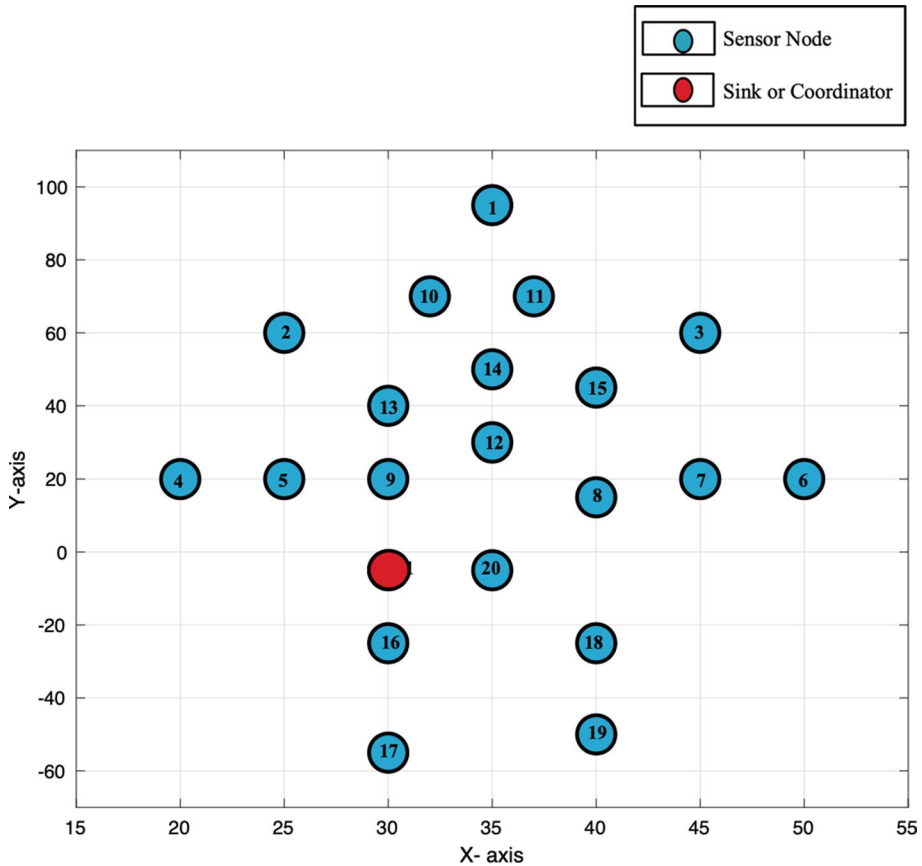


Fig. 9 Node deployment in MATLAB area (20 nodes)

send their data there. Other nodes send their data in multi-hop. Most evidently nodes 1 to 5 are positioned at upper half of the body and their fixed threshold based performances are better than incremental model and direct transmission protocol. In contrast nodes 6 to 10 are positioned in lower half of the body and fixed threshold based protocol performance is same as that of direct transfer protocol and shortest path protocol whereas incremental approach performs better. However overall energy consumption in incremental threshold scenario is less in comparison to other three scenarios.

Figure 11 shows that nodes send data to the sink in the same manner. Here nodes 1 to 5 are placed in the upper portion of the body which is at distant from the sink, thus fixed threshold based protocol performs better than the other three protocols. But, in case of nodes at lower part of the body, node numbers 7 to 10 are nearer to the sink therefore they send data to the sink in single hop. On the other hand, in incremental model only nodes 8 and 10 transfer data to the sink in single hop and other nodes transmit data to the sink in multi hop. It is seen that performance of incremental protocol is better than the other proposed protocols.

In the same simulation environment, number of nodes are increased to 15 but sink position (30, - 5) is fixed as in the previous scenario and protocols send data to the

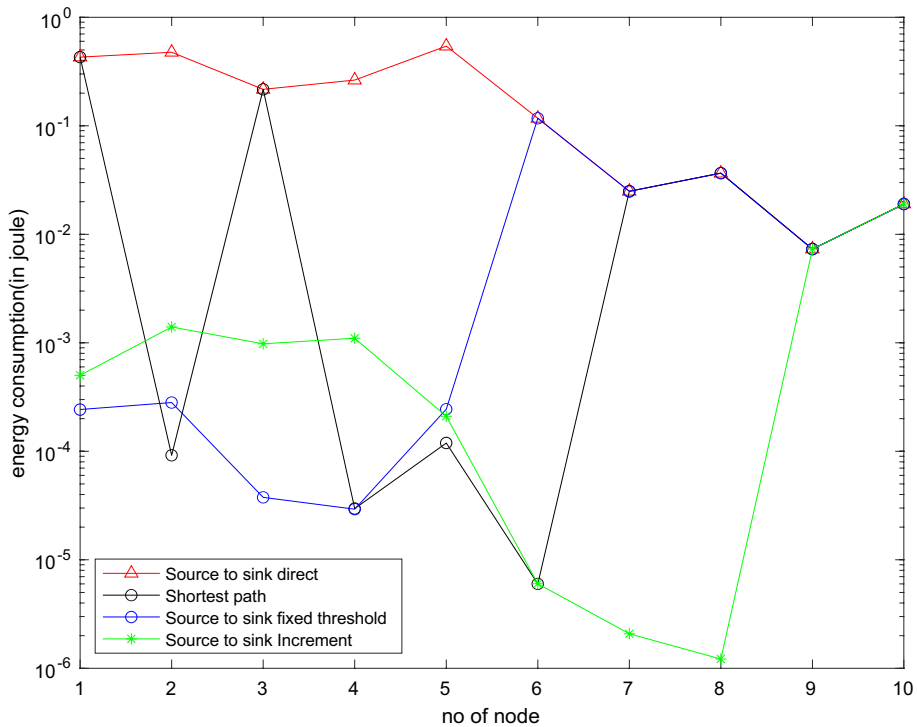


Fig. 10 Energy consumption of fixed position network configuration. (Color figure online)

sink either through single hop or multihop, depending on the distance multi hops vary. Two proposed protocols, fixed threshold model and Incremental model with source to sink and shortest path algorithm are applied in this environment. We have observed that when number of nodes increases i.e. in upper portion of the body surface, fixed threshold performs better but over all efficiency of incremental model is superior compared to the other three protocols shown in Fig. 12.

In Fig. 13, number of sensor nodes increases to 20 and these are placed randomly in the human body surface area. It is observed that when number of forwarder nodes is more over a certain range, here 0.5 cm (because corresponding SAR value of this transmission range is not harmful for human tissue) then Fixed threshold achieves better results but all-inclusive performance of the incremental protocol is more efficient compared to other three protocols.

In Fig. 14 shows that comparison graph of average energy consumption of the different protocol in different network configuration. We observed that each configuration fixed threshold model have less amount of energy consumption compared to the source to sink and shortest path algorithm even if increase the number of the node deployment. But in case of Incremental model, energy consumption is less compare to the threshold based routing as well as other two pre-defined protocols. Energy consumption of the incremental model is 85%, 84%, 98% and 23% less than the threshold based model in Network Configuration 1, Network Configuration 2, Network Configuration 3 and Network Configuration

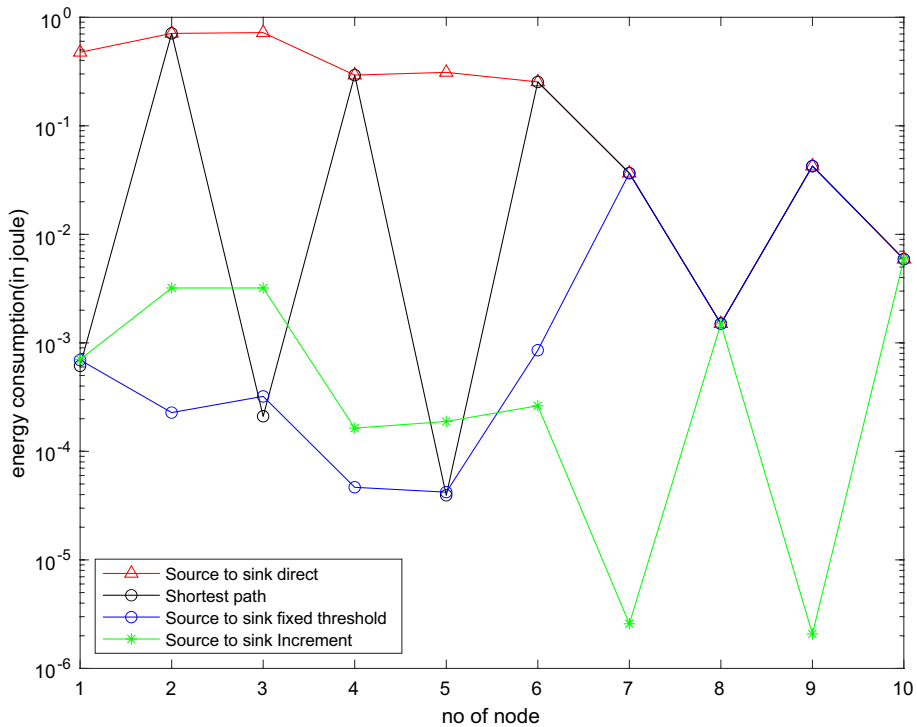


Fig. 11 Energy consumption of random position network configuration. (Color figure online)

4 respectively. Hence it is clearly seen that incremental model has less energy consumption compared to all network configurations with varying scenarios. So incremental model is efficient routing protocol to ensure reduced SAR.

7.2 Path Loss Performance Metric

Signal attenuation during transmission is represented by path loss. It is measured in decibel (dB). It is the difference between the transmitted power and received power with or without considering antenna gain. Path loss is inversely proportional to the surface area of propagating wave front, if it increases then path loss decreases and vice versa. It is calculated using distance between source node to sink and frequency.

Path-loss of the different network configurations are shown in Figs. 15, 16, 17, 18.

In WBAN scenario, different sensors are placed in fixed positions of the human body and with varying movements and postures, the transmitted signals are affected. Distance and frequency are major parameters to define path loss as expressed in Eq. 12 and frequency and path loss relationship is shown in Eq. 13.

$$PL(f, d) = PL(f) \times PL(d) \quad (12)$$

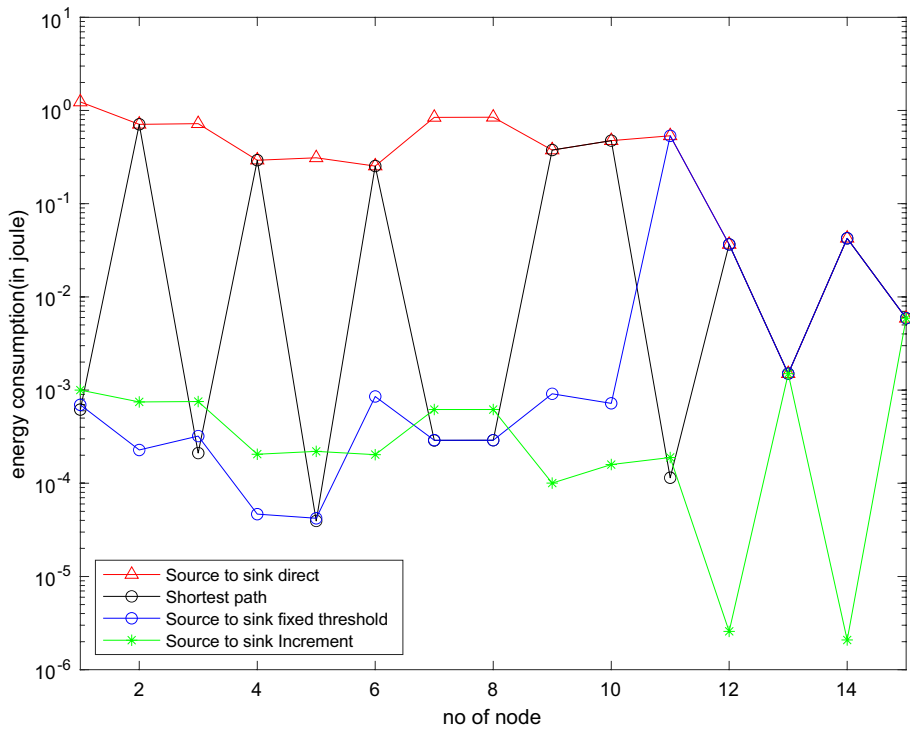


Fig. 12 Energy consumption of random position with 15 nodes network configuration. (Color figure online)

$$\sqrt{PL(f)} \propto f^m \quad (13)$$

where m denotes frequency dependent factor and it is related to the geometry of the body. The relation between distance and path loss is shown in Eq. (14) based on Friis formula.

$$PL(f, d) = PL_o + 10n \log_{10} \frac{d}{d_r} + X_\sigma \quad (14)$$

where PL is received power, d is the distance between transmitter and receiver, d_r is the reference distance, dependent propagation environment path loss coefficient is represented by n . Its value for free space is 2, for WBAN this value varies from 3 to 4 i.e. Line of Sight (LOS) communication and from 5 to 7.4 is for Non-line of sight (NLOS) communication. X represents Gaussian random value and σ is standard deviation. Here reference distance (d_r) value is 10 cm. X_σ is strength of the signal between receiver and transmitter boundary but it is difficult to consider. Here value used is 4.1. In all the scenarios constant frequency of 2.4 GHz is used, n is the path loss co-efficient in Eq. 14 with value 3.38. At reference distance d_r reference power is represented by PL_o .

PL_o is expressed as in Eq. 15 below:

$$PL_o = 10 \log_{10} \frac{(4 \times d \times f)^2}{c} \quad (15)$$

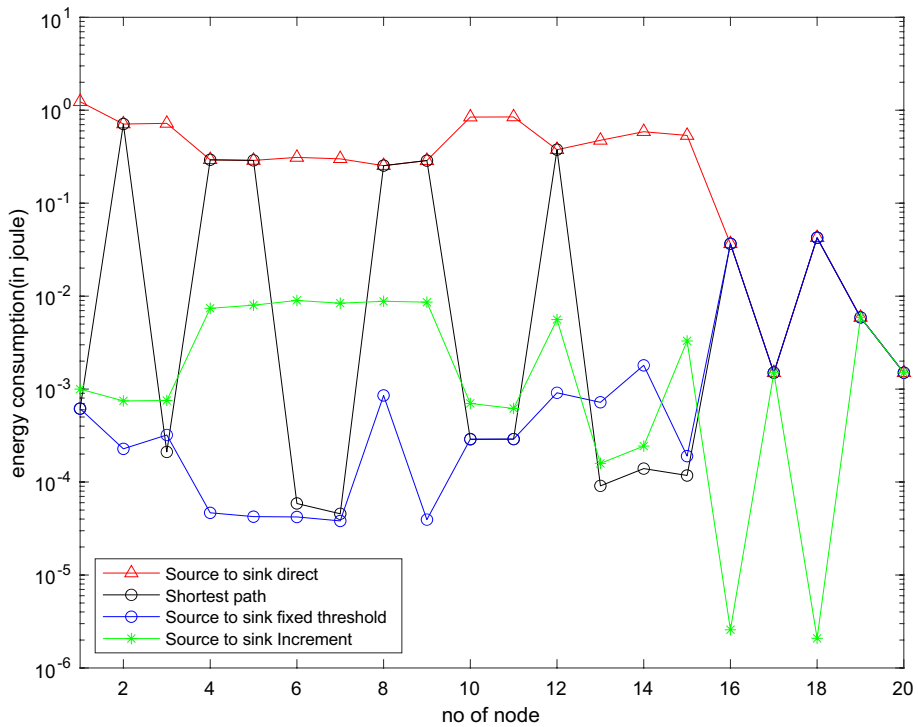


Fig. 13 Energy consumption of random position with 20 nodes network configuration. (Color figure online)

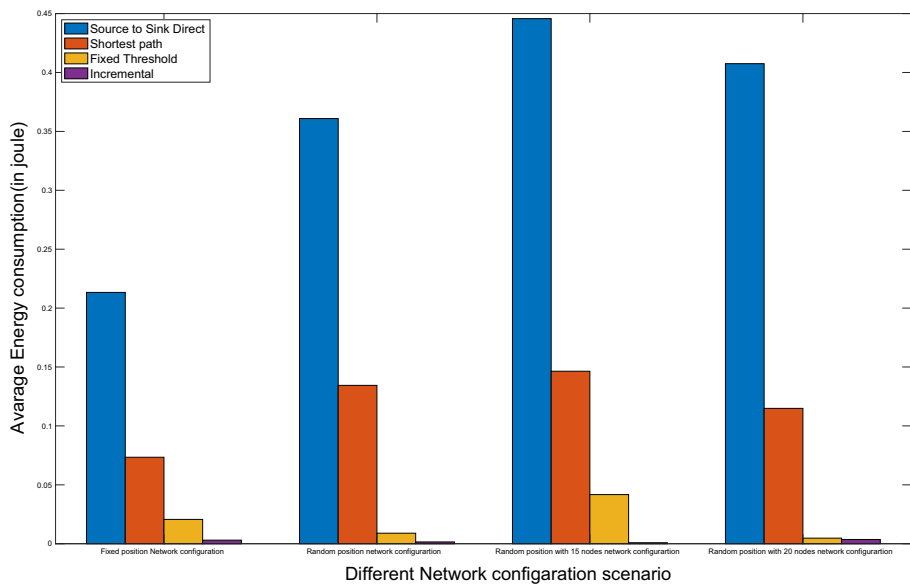


Fig. 14 Average energy comparison of different scenario

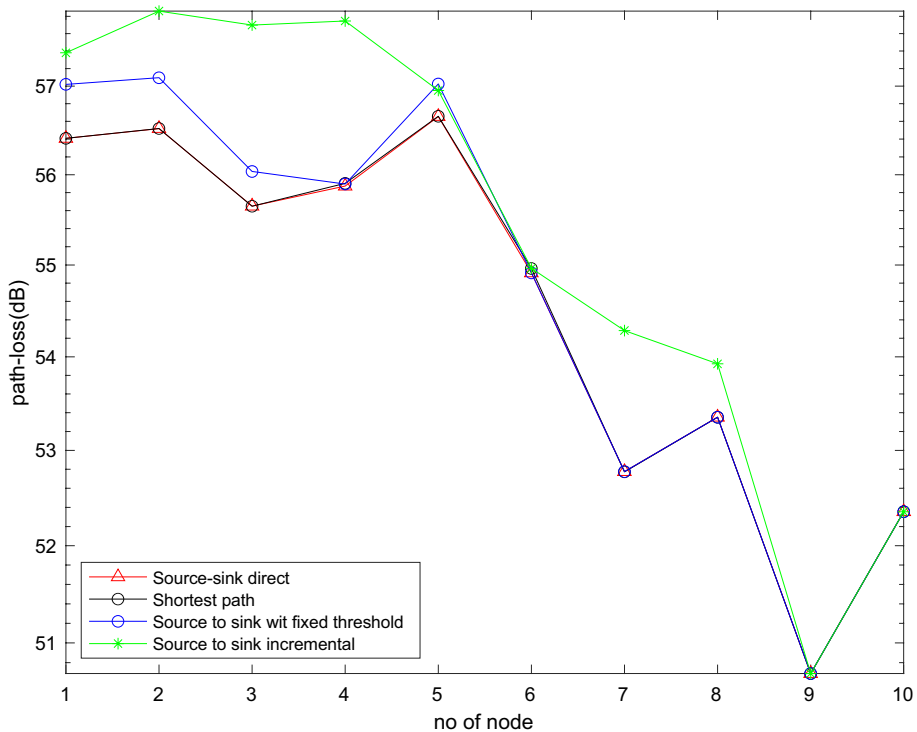


Fig. 15 Path-loss of fixed position network configuration. (Color figure online)

where c =speed of light, f =frequency of the communication signal and d =distance between transmitter and receiver in Eq. 15. Path loss is a function of distance and frequency. This can be inferred from Figs. 15, 16, 17, 18 which show the different path loss with sensor node placement in multiple scenarios and in each case, path loss in source to sink fixed threshold model is lowest due to Line of sight communication (LOS) having lowest surface area. In other models, data gets transmitted in multihop with NLOS communication. Here red, black, blue and green colour indicates source to sink direct, shortest path algorithm, fixed threshold based model and incremental model respectively.

In Fig. 15, proposed scenarios are compared with direct source to sink scenario and shortest path algorithm. In case of source to sink fixed threshold based model and incremental model path loss is more due to data transfer in source to sink multi hop communication. In both proposed protocol cases network stability is more compared to the source to sink direct and shortest path. However incremental threshold model has more path loss and stability compared to fixed threshold model.

In Fig. 16, path loss of the incremental proposed protocol is more with higher stability, same as in Fig. 15. In this network configuration node 8 has lesser path loss compared to other node positions because distance from node 8 to sink position is minimum and all protocols send data using LOS communication.

In Fig. 17, Number of nodes increases to 15 for see the path loss variation, We see that in the upper surface of the body i.e. upto node 11 deployment, path loss is minimum for

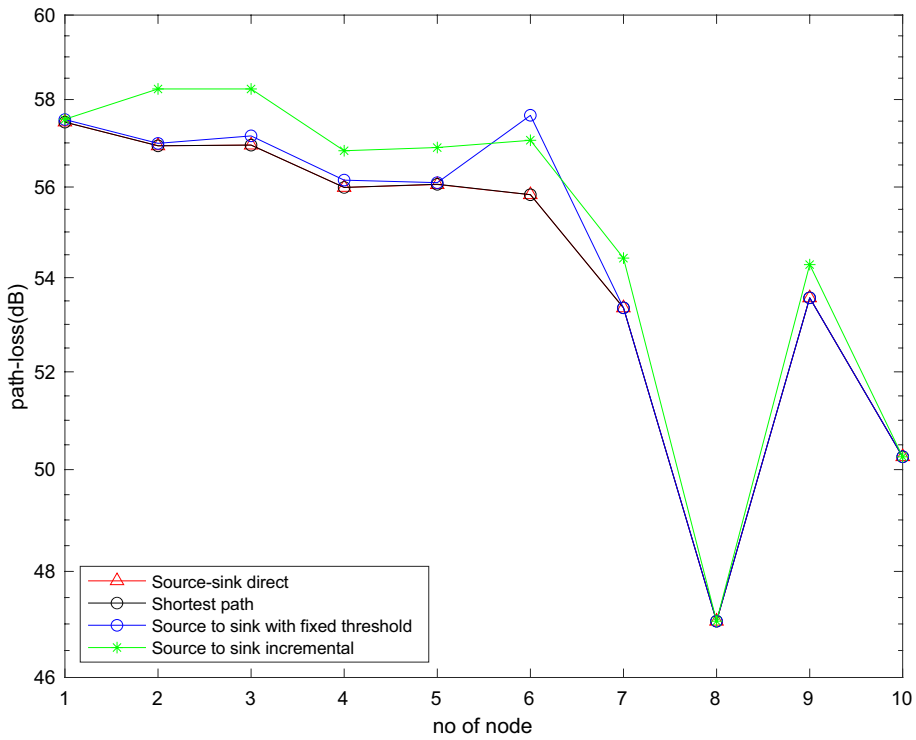


Fig. 16 Path-loss of random position network configuration. (Color figure online)

source to sink protocol and maximum incremental model but some cases fixed threshold is more due to distance and NLOS communication from source to sink. But in the lower portion of the body i.e. 12 to 15 maximum node send data to the sink using LOS communication. So path loss approximately same for all protocol. Moreover incremental model path loss is more compare to all protocols and it indicates that incremental model have more stability than other protocols.

Now, we increase the nodes to 20 in the simulation environment and path loss as shown in Fig. 18. In the upper portion of the body path loss is zigzag manner but with more cases it is concluded that incremental model path loss is higher compare to the rest. Only nodes 13 and 14 implement fixed threshold model where due to NLOS communication greater distance is covered. But in lower surface of the body all protocols show approximately same path-loss. Here, nodes 16 and 20 cover same distance from sink and they are using LOS communication. Henceforth, path loss of these two nodes throughout all protocols are same. However, general path loss of the incremental protocol is greater compared to others along with greater stability.

Figure 19 shows that comparison graph of total path loss of the different network configurations and different data transmission protocols. Incremental model have higher path loss compared to the other protocols.

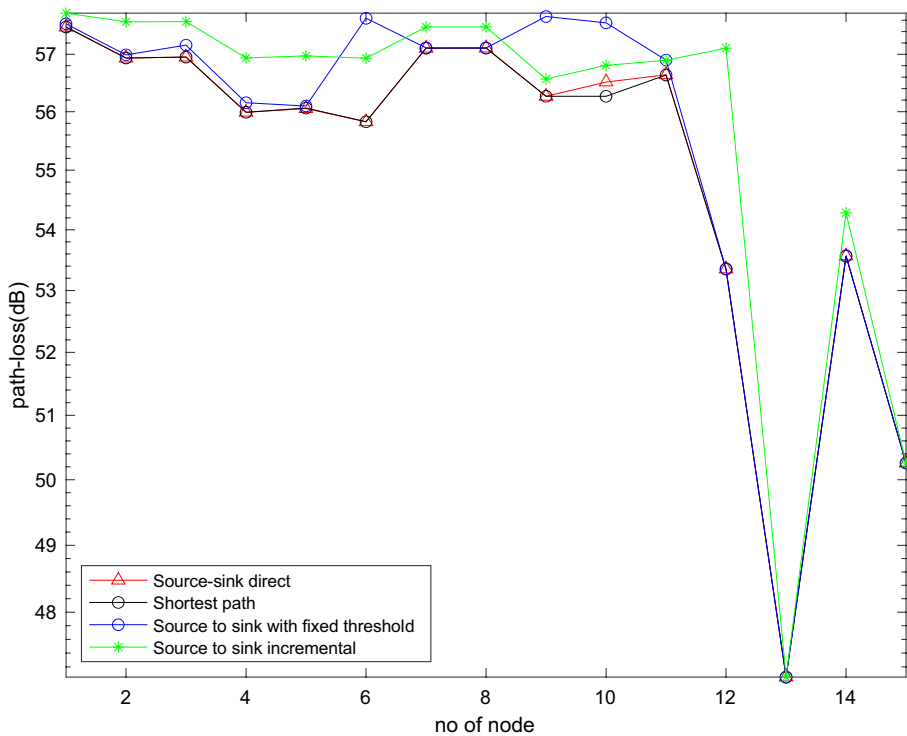


Fig. 17 Path-loss of random position with 15 nodes network configuration. (Color figure online)

7.3 Network Lifetime

Network lifetime is the total network operation time of data transmission. In WBAN nodes may run out of energy for data transmission due to drainage of the same during data transmission over long distance or forwarding other nodes' data [44, 45]. Also to forward repetitive or redundant sensor data may also be a pivotal cause of such unnecessary energy drainage and low network lifetime [45]. After data transmission in WBAN, the energy left is termed as wasted energy [46]. We can calculate network lifetime by using Eq. (16).

$$EL(i) = \left(E_i - \frac{E_w}{E_{Tx}} \right) \quad (16)$$

where E_i is the initial energy of the node and its value is 5 J. E_w represents the wasted energy after data transmission from source to sink. E_{Tx} is total energy consumption during transmission. In WBAN, it is constant at 1.67 J as per the energy model used.

Here in Figs. 20, 21, 22, 23, network lifetime of different network configurations are plotted with different conditions. The red line indicates Source to sink direct data transfer, black line indicates shortest path algorithm where as blue and green lines indicate source to sink fixed threshold and incremental protocol respectively.

Figure 20 shows that nodes 2 and 5 in source to sink direct protocol gives less network life time because these are at distant and transmission energy consumed is more compared to other protocols. In case of shortest path some sensor node sends data in single hop which

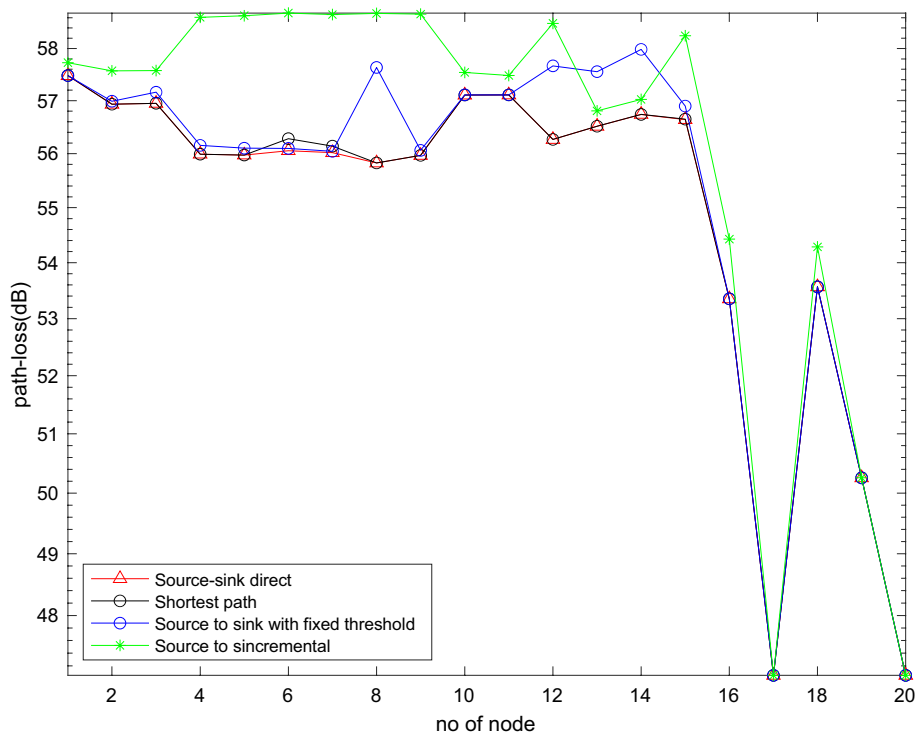


Fig. 18 Path-loss of random position with 20 nodes network configuration. (Color figure online)

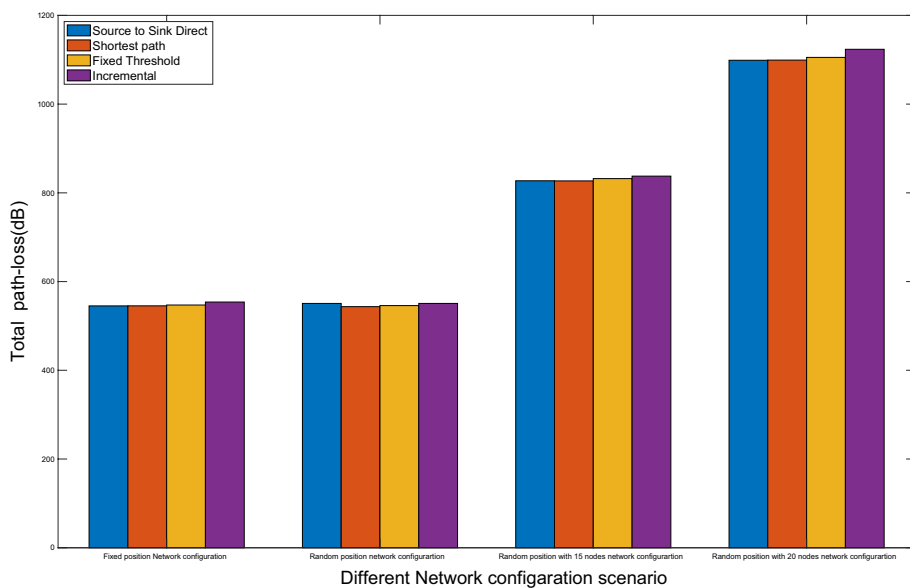


Fig. 19 Total path loss of different configuration and different scenario

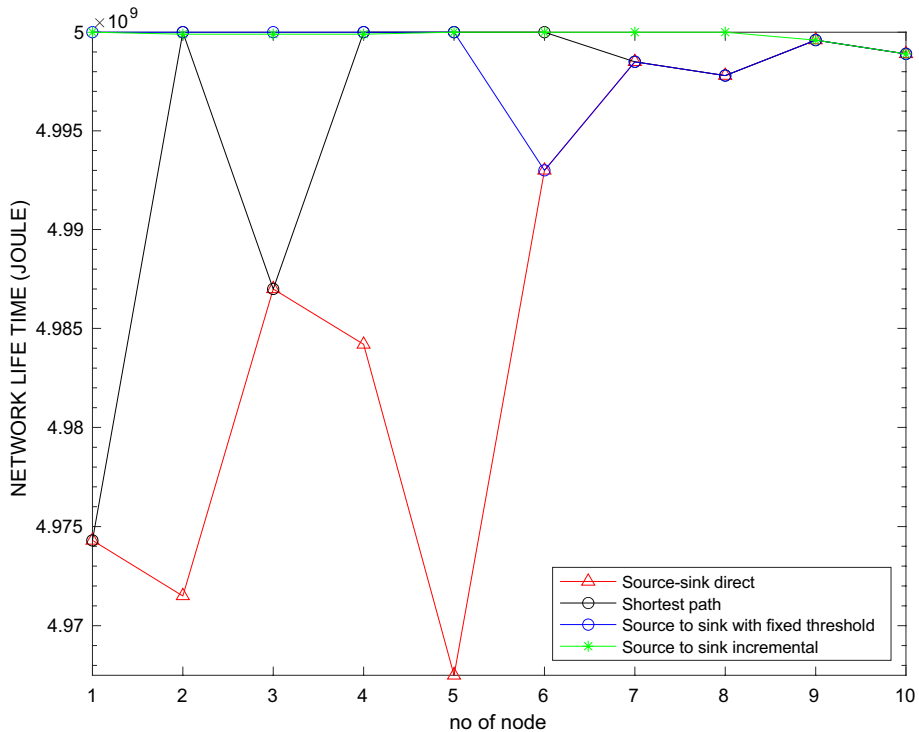


Fig. 20 Network lifetime for fixed network configuration (NC1). (Color figure online)

is closer to the sink while others send data in multi hop which are at greater distances from source to sink, here, node 1, 3, 5 and 6 use multi hop thus, less energy is consumed compared to source to sink direct transmission and approximately same as other two protocols. If we note that for fixed threshold model upto node 5 results are same as that of incremental protocol, however after that within range (here 0.5 cm because corresponding SAR value of this transmission range is not harmful for human tissue) it gets sink node, where data is sent directly to the sink as in source to sink direct protocol. It is clearly observed that in Fig. 20 fixed threshold model or incremental based model have higher network life time in comparison to the other two predefined protocols.

Figure 21 shows the network lifetime where random deployment of the 10 sensor nodes is achieved. Similar to Fig. 20, source to sink and shortest path network life time is less in equation to the other proposed protocols as their nodes consume more energy. Here Fixed threshold model fluctuate in nodes 6, 7, 8 and 9 because multi hop distance is lesser than that of the incremental model. Otherwise network life time of threshold based model is the same as incremental protocol. Therefore, it is clearly understood that network life time of network incremental protocol is higher than the other three protocols.

Again we increase the number of nodes to 15 and Fig. 22 shows its network life time. Here we see that similar to previous scenarios, depending upon the distance from source to sink and shortest path nodes more energy is consumed. In fixed threshold model for nodes 11, 12 and 14, more energy consumption is witnessed compared to incremental model so

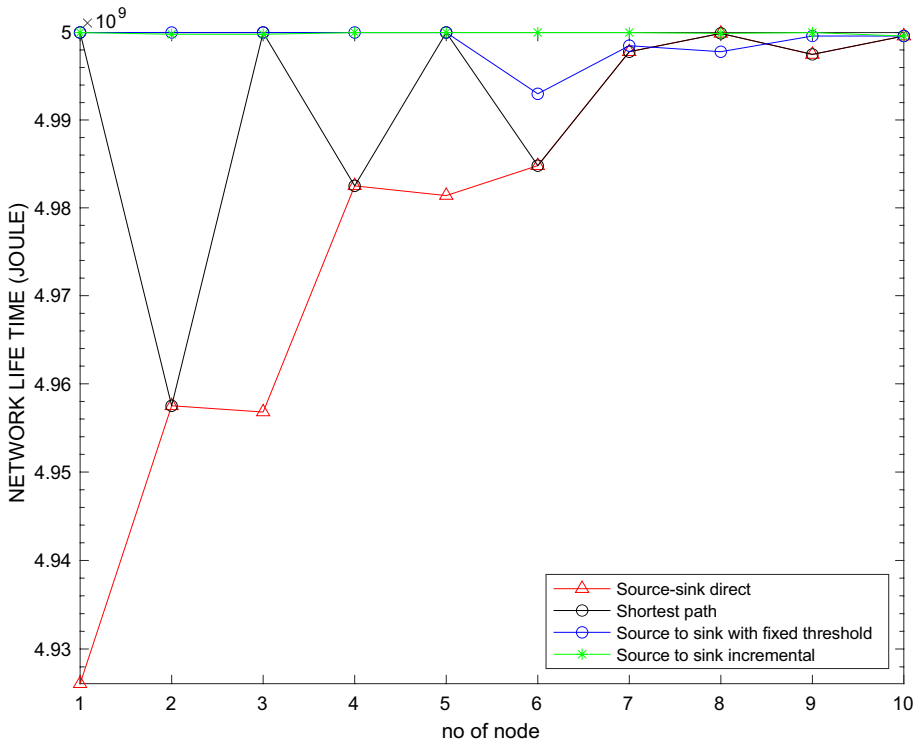


Fig. 21 Network-lifetime of random position network configuration (NC2-SC1: 10 nodes). (Color figure online)

network lifetime is less and for other nodes results are approximately same. It is clearly observed that in summary network life time is better in incremental model.

Figure 23 describes network life time with 20 nodes in the network configuration with random node positions (NC2-SC3). Another observation is that when number of nodes increases in a body surface area then both threshold based and incremental model act in a similar manner and network life time is high compared to other predefined protocols. It is clearly depicting that fixed threshold or incremental threshold based models have higher network life time in all the network configurations. It is also shown that in comparison incremental threshold based model is more efficient and has longer stability period.

8 Conclusion

The work proposed and implemented here is to focus on energy efficient routing techniques in WBAN attached to human body for efficient health monitoring. Performance evaluation of proposed routing algorithms for health data transmission from source to sink is done. Implementation results using MATLAB shows routing from source to sink

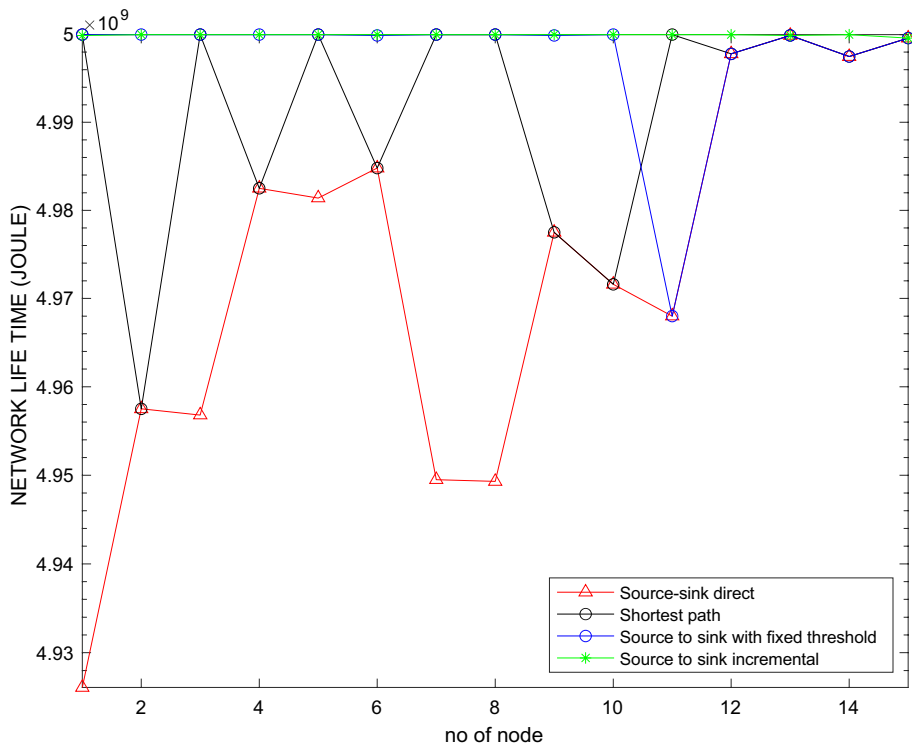


Fig. 22 Network-lifetime of random position with 15 nodes network configuration ((NC2–SC2: 15 nodes). (Color figure online)

using incremental distance performs better compared to direct single hop routing or fixed threshold based in-between distance approach in terms of energy consumption, path loss and network lifetime for same network configuration. Incremental distance based routing keeps SAR low as transmission power required in this approach is low thus can avoid health hazards. For rigorous performance evaluation of proposed data

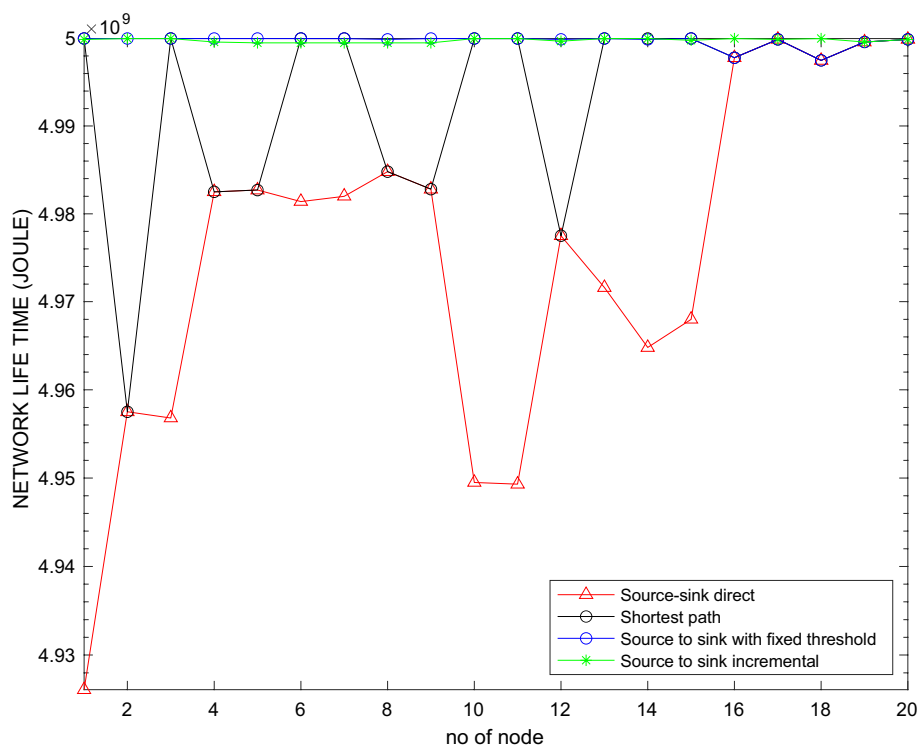


Fig. 23 Network-lifetime of random position with 20 nodes network configuration (NC2–SC3: 20 nodes). (Color figure online)

transmission protocols, several network configurations, scenarios and node deployment strategies i.e. fixed or random positioning have been defined and implemented.

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Data availability (Data transparency): Available.

Code availability (Software application or custom code): Available.

Compliance with Ethical Standards

Conflict of interest The authors declare that they have no conflict of interest.

References

1. Ha, I. (2016). Even energy consumption and backside routing: An improved routing protocol for effective data transmission in wireless body area networks. *International Journal of Distributed Sensor Networks*, 12(7), 1550147716657932.
2. Saha, R., Biswas, S., & Pradhan, G. (2017). A priority based routing protocol with extensive survey and comparison of related works for healthcare applications using WBAN. In *2017 International conference on wireless communications, signal processing and networking (WiSPNET)* (pp. 1424–1430). IEEE.
3. Biswas, S., Bhattacharya, T., & Saha, R. (2018). On fall detection using smartphone sensors. In *2018 International conference on wireless communications, signal processing and networking (WiSPNET)* (pp. 1–4). IEEE.
4. Patel, M., & Wang, J. (2010). Applications, challenges, and prospective in emerging body area networking technologies. *IEEE Wireless Communications*, 17(1), 80–88.
5. Qadri, Y. A., Nauman, A., Zikria, Y. B., Vasilakos, A. V., & Kim, S. W. (2020). The future of healthcare internet of things: A survey of emerging technologies. *IEEE Communications Surveys and Tutorials*, 22(2), 1121–1167.
6. Cavallari, R., Martelli, F., Rosini, R., Buratti, C., & Verdone, R. (2014). A survey on wireless body area networks: Technologies and design challenges. *IEEE Communications Surveys and Tutorials*, 16(3), 1635–1657.
7. Movassaghi, S., Abolhasan, M., Lipman, J., Smith, D., & Jamalipour, A. (2014). Wireless body area networks: A survey. *IEEE Communications Surveys and Tutorials*, 16(3), 1658–1686.
8. Rehmani, M. H., & Pathan, A. S. K. (Eds.). (2016). *Emerging communication technologies based on wireless sensor networks: current research and future applications*. Boca Raton: CRC Press.
9. Sharma, R., Ryait, H. S., & Gupta, A. K. (2016). Wireless body area network—a review. *Research Cell: An International Journal of Engineering Sciences*, 17, 494–499.
10. Qu, Y., Zheng, G., Ma, H., Wang, X., Ji, B., & Wu, H. (2019). A survey of routing protocols in WBAN for healthcare applications. *Sensors*, 19(7), 1638.
11. Agarwal, N., B. S., & Saif, S. R. (2019). Evaluation of WBAN with and without temporal model using Castalia”. *Tech Vistas, MAKAUT, WB*, 1(2), 10–21.
12. Javaid, N., Abbas, Z., Fareed, M. S., Khan, Z. A., & Alrajeh, N. (2013). M-ATTEMPT: A new energy-efficient routing protocol for wireless body area sensor networks. *Procedia Computer Science*, 19, 224–231.
13. Khan, Z. A., Sivakumar, S., Phillips, W., & Robertson, B. (2014). ZE-QoS: A new energy and QoS-aware routing protocol for communication of sensor devices in healthcare system. *International Journal of Distributed Sensor Networks*, 10(6), 627689.
14. Bangash, J. I., Abdullah, A. H., Anisi, M. H., & Khan, A. W. (2014). A survey of routing protocols in wireless body sensor networks. *Sensors*, 14(1), 1322–1357.
15. Aziz, Z., Qureshi, U. M., Shaikh, F. K., Bohra, N., Khelil, A., & Felemban, E. (2016). Revisiting routing in wireless body area networks. *Emerging Communication Technologies Based on Wireless Sensor Networks: Current Research and Future Applications*, 89.
16. Shahbazi, Z., & Byun, Y. C. (2020). Towards a secure thermal-energy aware routing protocol in wireless body area network based on blockchain technology. *Sensors*, 20(12), 3604.
17. Nabila, A. (2019). A QoS based comparative analysis of the IEEE standards 802.15. 4 & 802.15. 6 in WBAN-based healthcare monitoring systems. In *2019 International conference on wireless technologies, embedded and intelligent systems (WITS)* (pp. 1–5). IEEE.
18. Huynh, T. T., Dinh-Duc, A. V., & Tran, C. H. (2016). Delay-constrained energy-efficient cluster-based multi-hop routing in wireless sensor networks. *Journal of Communications and Networks*, 18(4), 580–588.
19. Rashed, M., Kabir, M. H., & Ullah, S. E. (2012). WEP: An energy efficient protocol for cluster based heterogeneous wireless sensor network. arXiv preprint <https://arxiv.org/abs/1207.3882>.
20. Ivanov, S., Foley, C., Balasubramaniam, S., & Botvich, D. (2012). Virtual groups for patient WBAN monitoring in medical environments. *IEEE Transactions on Biomedical Engineering*, 59(11), 3238–3246.
21. Monowar, M. M., Mehedi Hassan, M., Bajaber, F., Hamid, M. A., & Alamri, A. (2014). Thermal-aware multiconstrained intrabody QoS routing for wireless body area networks. *International Journal of Distributed Sensor Networks*, 10(3), 676312.
22. Ahmad, A., Javaid, N., Qasim, U., Ishfaq, M., Khan, Z. A., & Alghamdi, T. A. (2014). RE-ATTEMPT: A new energy-efficient routing protocol for wireless body area sensor networks. *International Journal of Distributed Sensor Networks*, 10(4), 464010.

23. Javaid, N., Ahmad, A., Nadeem, Q., Imran, M., & Haider, N. (2015). iM-SIMPLE: iMproved stable increased-throughput multi-hop link efficient routing protocol for wireless body area networks. *Computers in Human Behavior*, 51, 1003–1011.
24. Ul Huque, M. T. I., Munasinghe, K. S., & Jamalipour, A. (2014). Body node coordinator placement algorithms for wireless body area networks. *IEEE Internet of Things Journal*, 2(1), 94–102.
25. Adhikary, S., Choudhury, S., & Chattopadhyay, S. (2016). A new routing protocol for WBAN to enhance energy consumption and network lifetime. In *Proceedings of the 17th international conference on distributed computing and networking* (pp. 1–6).
26. Smail, O., Kerrar, A., Zetili, Y., & Cousin, B. (2016). ESR: Energy aware and stable routing protocol for WBAN networks. In *2016 International wireless communications and mobile computing conference (IWCMC)* (pp. 452–457). IEEE.
27. Ahmed, G., Jianhua, Z., & Fareed, M. M. S. (2017). PERA: priority-based energy-efficient routing algorithm for WBANs. *Wireless Personal Communications*, 96(3), 4737–4753.
28. Khan, R. A., Mohammadani, K. H., Soomro, A. A., Hussain, J., Khan, S., Arain, T. H., & Zafar, H. (2018). An energy efficient routing protocol for wireless body area sensornetworks. *Wireless Personal Communications*, 99(4), 1443–1454.
29. Saha, R., Naskar, S., Biswas, S., & Saif, S. (2019). Performance evaluation of energy efficient routing with or without relay in medical body sensor network. *Health and Technology*, 9(5), 805–815.
30. Ullah, F., et al. (2019). Traffic priority based delay-aware and energy efficient path allocation routing protocol for wireless body area network. *Journal of Ambient Intelligence and Humanized Computing*, 10(10), 3775–3794.
31. Sagar, A. K., Singh, S., & Kumar, A. (2020). Energy-aware WBAN for health monitoring using critical data routing (CDR). *Wireless Personal Communications*, 112, 273–302.
32. Zhang, Y., Zhang, B., & Zhang, S. (2020). An adaptive energy-aware relay mechanism for IEEE 802.15.6 wireless body area networks. *Wireless Personal Communications*, 115, 2363–2389.
33. Navya, V., Deepalakshmi, P. (2018). Threshold-based energy-efficient routing for transmission of critical physiological parameters in a wireless body area network under emergency scenarios. *International Journal of Computers and Applications*, 1–10.
34. Tariq, F., Anwar, M., Janjua, A. R., Khan, M. H., Khan, A. U., & Javaid, N. (2020). Blockchain in WSNs, VANets, IoTs and healthcare: A survey. In *Workshops of the international conference on advanced information networking and applications* (pp. 267–279). Cham: Springer.
35. Li, Y., Bakaloglu, B., & Chakrabarti, C. (2007). A system level energy model and energy-quality evaluation for integrated transceiver front-ends. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 15(1), 90–103.
36. Bernardi, P., Cavignaro, M., Pisa, S., & Piuze, E. (2000). Specific absorption rate and temperature increases in the head of a cellular-phone user. *IEEE Transactions on Microwave Theory and Techniques*, 48(7), 1118–1126.
37. Jang, C.W. (2005). *U.S. Patent No. 6,920,344*. Washington, DC: U.S. Patent and Trademark Office.
38. Hochwald, B. M., & Love, D. J. (2012). Minimizing exposure to electromagnetic radiation in portable devices. In *2012 Information theory and applications workshop* (pp. 255–261). IEEE.
39. Mouzehkesh, N., Zia, T., Shafigh, S., & Zheng, L. (2015). Dynamic backoff scheduling of low data rate applications in wireless body area networks. *Wireless Networks*, 21(8), 2571–2592.
40. Sakurai, T., Kiyokawa, T., Narita, E., Suzuki, Y., Taki, M., & Miyakoshi, J. (2011). Analysis of gene expression in a human-derived glial cell line exposed to 2.45 GHz continuous radiofrequency electromagnetic fields. *Journal of Radiation Research*, 52(2), 185–192.
41. Kathe, K. S., & Deshpande, U. A. (2019). A thermal aware routing algorithm for a wireless body area network. *Wireless Personal Communications*, 105(4), 1353–1380.
42. Chiti, F., Fantacci, R., Archetti, F., Messina, E., & Toscani, D. (2009). An integrated communications framework for context aware continuous monitoring with body sensor networks. *IEEE Journal on Selected Areas in Communications*, 27(4), 379–386.
43. Semiconductor, N. (2007). nRF24L01 Single Chip 2.4 GHz Transceiver, (Product Specification Revision 2.0).
44. Mathew, R., Younis, M., & Elsharkawy, S. M. (2005). Energy-efficient bootstrapping for wireless sensor networks. *Innovations in Systems and Software Engineering*, 1(2), 205–220.
45. Mondal, A. S., Neogy, S., Mukherjee, N., & Chattopadhyay, S. (2019). A survey of issues and solutions of health data management systems. *Innovations in Systems and Software Engineering*, 15(2), 155–166.
46. Himsoon, T., Siriwoongpairat, W. P., Han, Z., & Liu, K. R. (2007). Lifetime maximization via cooperative nodes and relay deployment in wireless networks. *IEEE Journal on Selected Areas in Communications*, 25(2), 306–317.

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